

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Winter 2023

Course: B. Tech.

Branch : E & T C

Semester : V

Subject Code & Name: BTETC503 - Analog Communication Engineering

Max Marks: 60

Date:

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
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(Level/CO) Marks

Q.1 Answer Any Two of the following.

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|--|-----------------|----------|
| A) Define Communication and Explain the complete Analog communication system. | CO 01 | 6 |
| B) What is modulation? Give the classification Modulation with definition | CO 01 | 6 |
| C) Write the difference between i)FDM and TDM ii) Analog communication and Digital communication | CO 01,07 | 6 |

Q.2 Answer Any Two of the following.

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|--|--------------|----------|
| A) Explain DSBFC Technique in detail with the help of (i) frequency spectrum (ii) Time domain representation (iii) power relation with carrier wave | CO 02 | 6 |
| B) Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to the depth of (i) 70% and (ii) 65% | CO 02 | 6 |
| C) Explain generation of AM using Nonlinear resistance device | CO 03 | 6 |

Q.3 Answer/Solve Any Two of the following.

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|---|--------------|----------|
| A) What is Angle modulation? Derive the mathematical expression for PM. | CO 05 | 6 |
| B) Explain Narrowband and Wideband FM Also compare the FM and AM. | CO 05 | 6 |
| C) In an FM system, when the audio frequency (AF) is 300 Hz, and the AF voltage is 1.2 V, the deviation is 2.4 kHz. If the AF voltage is now increased to 3.6 V, what is the new deviation? If the AF voltage is further raised to 10 V while the AF is dropped to 150 Hz, what is the deviation? Find the modulation index in each case. | CO 05 | 6 |

Q.4 Answer the following.

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|--|--------------|----------|
| A) Define:
(i) Sensitivity, (ii) Selectivity, (iii) Fidelity, (iv) Image frequency and its rejection. | CO 07 | 6 |
| B) Explain working of Superhetrodyne receiver in detail. | CO 04 | 6 |

Q. 5 Solve the following.

- A)** Two resistors $20\text{ k}\Omega$ and $50\text{ k}\Omega$ are at room temperature (290 K). Calculate for bandwidth of 100 KHz, the thermal noise for the following conditions: **CO 04 6**
- (i) For resistor $20\text{ k}\Omega$
 - (ii) For resistor $50\text{ k}\Omega$
 - (iii) For two resistors in series
 - (iv) For two resistors in parallel.
- B)** An amplifier operating over the frequency range from 3 to 10MHz has a 20K input resistance. What is the rms noise voltage at the input to this amplifier at room temperature? **CO 04, 06 6**

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